X.509 certificates. It is supported by all native clients like Linux, Windows, Mac OS X, FreeBSD and Blackberry OS.

Features

- strongSwan supports Elliptic Curve Cryptography (ECDH groups and ECDSA certificates and signatures) both for IKEv2 and IKEv1.
- Automatic assignment of virtual IP addresses to VPN clients from one or several address pools using either the IKEv1 ModeConfig or IKEv2 configuration payload.
- strongSwan IKEv2 NetworkManager applet supports EAP, X.509 certificate and PKCS#11 smartcard-based authentication.
- Modular plugins for crypto-algorithms and relational database interfaces.
- Automatic insertion and deletion of IPsec-policy based firewall rules.
- Dynamic IP address and interface update with IKEv2 MOBIKE.
- Fully tested support of IPv6 IPsec tunnel and transport connections.

Official website: *https://www.strongswan.org/*
Latest version: 5.7.1

WireGuard

WireGuard is a multi-platform, open source, simple, fast and most easy-to-use VPN solution available for sysadmins. It uses modern cryptography. It is better than IPSec and OpenVPN in terms of configuration, performance and usability. It is easy to use like SSH, and a VPN connection is developed by simply exchanging public keys, while everything else is handled by WireGuard.

WireGuard uses state-of-the-art cryptography, like the Noise Protocol Framework, Curve25519, ChaCha20, Poly1305, BLAKE2, SipHash24, HKDF, and secure trusted constructions.

WireGuard implements VPN techniques to create secure point-to-point connections in routed or bridged configurations, and it runs as a module inside the Linux kernel. A process called cryptokey routing is at the heart of WireGuard encryption. The mechanism works by associating public encryption keys with a list of VPN tunnel IP addresses that are allowed inside the tunnel. A unique private key and a list of peers are associated with each network interface. Each of the peers has a short and simple public key, used to authenticate it with other peers. These public keys may be distributed for use in configuration files in a number of ways, much like the transmission of SSH public keys.

In any server configuration, each peer (client application, etc) can send packets to the network interface with a source IP address that matches its corresponding list of permitted IP addresses. When the network interface wishes to send a packet to a peer, it looks at the destination IP of the data packet, and compares it to each peer's list of permitted IPs, in order to determine which peer to send it to.

Features

- WireGuard is designed as a general-purpose VPN for running on embedded interfaces and super computers alike, fit for many different circumstances.
- It is currently under heavy development, but already might be regarded as the most secure, easiest to use, and simplest VPN solution in the industry.
- Though it is still under development, even in its unoptimised state, it is faster than the popular OpenVPN protocols and delivers comparatively lower ping times.

Official website: *https://www.wireguard.com/*
Latest version: 0.0.20181006

FreeLan

FreeLan is an open source, multi-platform, peer-to-peer VPN software with no GUI but is packed with a multitude of features that allow users to surf the Web anonymously. It implements a peer-to-peer, full mesh VPN to create secure site-to-site and point-to-point connections in routing or bridged mode.

FreeLan uses the Open Secure Sockets Layer (OpenSSL) library to provide encryption of both the data and control channels. It can be configured in the following three ways.

- *Client-server:* With one computer acting as a server and the rest as clients. The server can also decide whether the clients can communicate with each other or not. This configuration is considered fragile because if anything happens to the server, the complete network can be destroyed.
- *Peer-to-peer:* In peer-to-peer network configuration, each node (device) is connected to all other nodes and if any node is disconnected, the entire network is not disturbed; so connectivity is ensured at all times.
- *Hybrid:* This is a combination of the client-server and peer-to-peer networks.

Features

- Offers pre-shared keys, certificate-based and user name-password based authentication.
- Written in C++ and available under the GNU GPL.
- Generic VPN software, not a Web proxy service.

Official website: *https://www.freelan.org/*
Latest version: 2.0.0 END

References

[1] *https://www.softether.org/*
[2] *https://github.com/trailofbits/algo*
[3] *https://github.com/StreisandEffect/streisand*
[4] *https://openvpn.net/*
[5] *https://pritunl.com/*
[6] *https://www.strongswan.org/*
[7] *https://www.wireguard.com/*
[8] *https://www.freelan.org/*

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